



RACINE2.2: A Software Application for Processing and Mapping Spatial Distribution of Root Length Density and Potential Root Extraction Ratio from Root Counts on Trench Profiles

Awa Faye, Laurent Laplaze, Bassirou Sine, and Jean-Louis Chopart

Abstract

A method has been developed to measure root intersection density (RID) on a trench-profile in field conditions. Here we describe how 2D spatial distribution mapping of RID can be processed and converted into root length density (RLD) and root distances (ARD) using a new freeware named RACINE2.2. The software also allows a simple modeling of potential root extraction ratio in the soil (PRER). The software contains models calculating RLD, ARD, and PRER from RID for several crops (maize, sorghum, sugarcane, rice, pearl millet, pineapple, eucalyptus). Models may be changed or added into RACINE2.2. RLD, ARD, and PRER are calculated for each spatial unit and can be used to generate 2D maps using RACINE2.2. Data can be exported to a spreadsheet or a surface mapping software for further analysis. It is also possible to import data into RACINE2.2 from a spreadsheet. This application thus makes studies about root–soil interactions, root growth, and root uptake easier. It opens new avenues to characterize root systems to improve root water and nutrient uptake in field conditions.

Key words Trench-profile method, Root profile in field conditions, Root length density, Average root distances, Potential root extraction in soil, Root mapping, Modeling, Freeware

1 Introduction

RACINE2.2 is an application developed to process root data from field root data obtained using the trench profile method [1–3], measuring and mapping the density of intersections between the roots and a plane (the soil profile). RACINE2.2 then makes it possible to calculate different functional characteristics of the root system, in small soil volumes namely root length density (RLD), average root distance (ARD), and potential root extraction ratio in soil (PRER) thanks to adapted models included in the application or added by the user. It is finally possible to map the spatial distribution of these root traits. This approach makes it possible

to perform agronomic diagnostics (effects of soil, climate, cropping practices) or cultivar comparisons on root system characteristics in experimental stations or in farmers' fields [2–4]. Taking into account spatial variability of root system in relation with soil characteristics may be very important for root water and nutrient uptake in field conditions. Consequently, this tool can contribute to improved modeling of soil–root and shoot–root interactions.

2 Materials

To use RACINE2.2, spatialized root counts on a soil profile are required, using a method for quantifying the number of intersections (NI) between a plane (soil profile) and roots [2, 3]. Measurements must be made on a vertical plane that is perpendicular to the row for row crops. RACINE2.2 includes models to calculate RLD, ARD, and PRER from such data. It makes it possible to move (1) from measured root densities on a surface to root lengths densities (RLD), (2) from RLD to average root distances (ARD), and (3) from ARD to potential root extraction ratio (PRER). These models are taken from the bibliography (*see Note 1* for more information). RACINE 2.2 was developed with Excel (Microsoft Excel 2013). The corresponding Microsoft Excel files (about 5 MB) in English or French language, with user guide in both French and English, are available for downloading at <http://www.diade.ird.fr/en/resources/209-racinesuk.html>. When it is open, RACINE2.2 automatically opens a folder called “Racine2Excel” on the computer’s hard disk. All folders opened by RACINE2.2 will be backed up in this space.

3 Methods

3.1 Getting Started with RACINE2.2

The home page contains, at the first start, a warning “Protected mode.” Click on the top button in the yellow “Enable Editing” (Fig. 1), then, still in the same yellow frame, click “Enable Content.” This procedure is only necessary at first use. The software is now functional and has a homepage with functional buttons (Fig. 1).

When starting for the first time on a workstation, error messages may appear depending on its configuration (*see Note 2*). RACINE2.2 does not always open on the home page (*see Note 3*). The home page (Fig. 1) briefly describes the software. It contains several buttons. The first two are (1) “Help” to open the user manual and (2) “More information” that gives access to general information about the application. The others are used to (1): “Import and export from RACINE2.2,” (2) “Exit” to exit the application and (3) “Open or delete an existing file,” to open

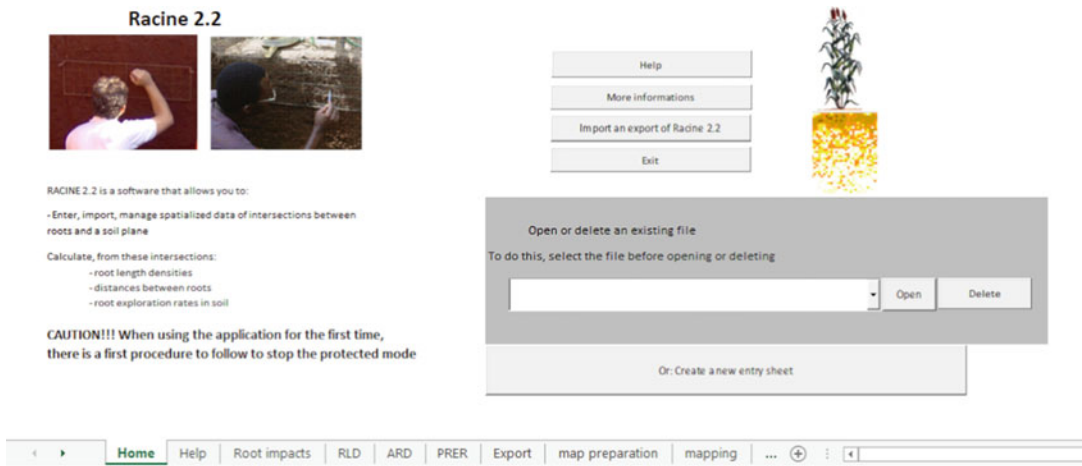


Fig. 1 Home page of RACINE2.2 during the first start and subsequent openings

or delete an existing file. To work on it, click on the small black arrow to display and select the selected work file (*see Note 4* for details), (4) “Create a new sheet” to add a new file with a new entry sheet. There are therefore two possibilities to start managing data with RACINE2.2: “Create a new file” and “Open an existing file.”

3.2 Creating a New File

To generate a new entry sheet, click on the button: “Create a new entry sheet” located at the bottom of the home page. A new window entitled “Enter the title of the entry” then appears. Enter the file title using only alphanumeric characters, spaces and dots. Then click on “OK” to generate the entry sheet (file). The tool then automatically positions itself in the “Input” tab with a page for “initializing the tables” (Fig. 2).

To generate these tables, it is necessary to provide information to define the size of the input tables:

1. Width and depth: enter the width and depth of the measured soil profile (cm).
2. Mesh: enter the length of the sides of the root count surface, always square (cm).
3. Repetitions: enter the number of profiles of identical dimensions measured.
4. Planting date and measurement date: write this information in the same format as the example. The table is prefilled. It is easy to modify the values (*see Note 5*). On the right there are two buttons to develop and save tables. When these basic data about the file characteristics are entered, click on “Save data,” then on “Initialize tables.” This creates a page with empty input tables (Fig. 3) configured according to the information provided. There are three tables per repetition, one per root



Fig. 2 Page used to develop root count (NI) input tables. On the left: table of file characteristics

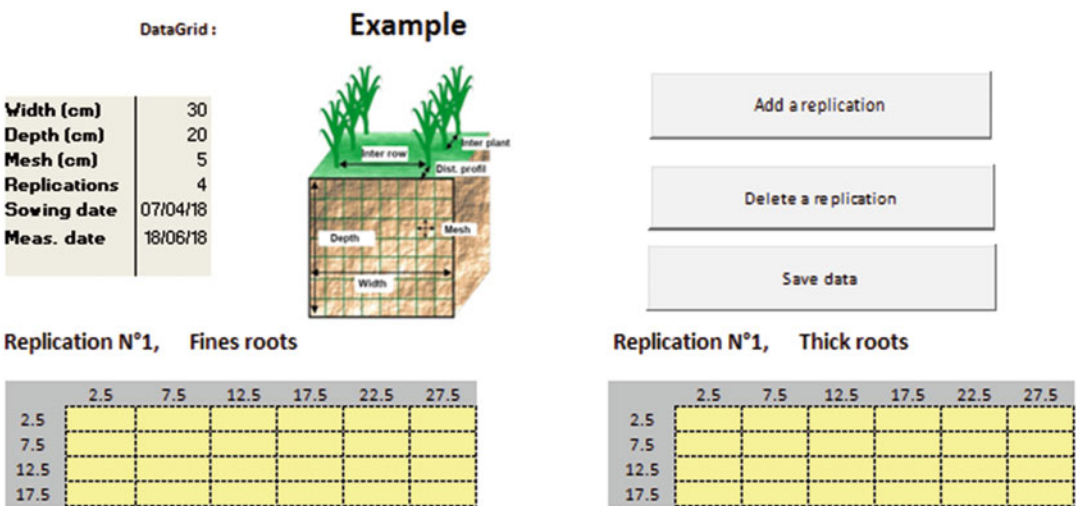


Fig. 3 Data input page appearing after creating the tables

type: thin, large, all roots (Fig. 3). Each mesh is located in the soil profile (horizontal and vertical position) and identified by its center. The first mesh is located to the top left of the table (*see Note 6*).

3.3 Working on an Existing File

In the home page (Fig. 1), use the “open or delete an existing file” button. To select an existing file in RACINE2.2, click on the black arrow. Then, click on the name of the working file and click on the “Open” button. The file, with the tables of root intersection numbers (NI) already entered, then appears (*see Note 4*) (Fig. 4). It is possible to delete a repetition (the last one) or add one, by clicking on the corresponding grey button at the top of the page (Fig. 4).

3.4 Entry or Import of the Number of Roots Intersecting a Soil Profile

When the tables of a file have been created (or modified or imported), these tables must be filled with the basic data from field experiments, that is, the number of roots intersection counted per mesh (NI). The NI input is made manually. It is suggested to frequently click on “Save data” to avoid losing the entries. It is always possible to modify the values entered. When tables

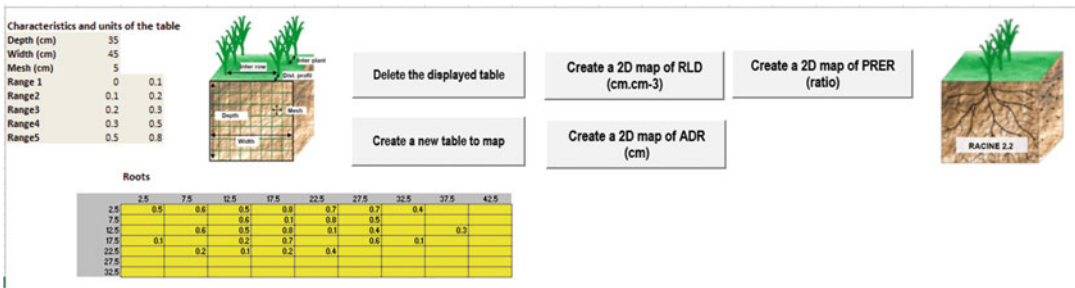


Fig. 4 Page displayed after clicking on the “Map preparation” tab

corresponding to “Fine roots” and “Thick roots” have been entered, it is easy to automatically fill in the table corresponding to “All roots” by the following procedure: copy, then special pasting (with the addition operation). It is also possible to import data from a spreadsheet or a field equipment. The interest is to (1) integrate data captured in the field into the application, (2) allow exchanges of basic data between users of RACINE2.2, (3) recalculate transformed values (e.g., averages). There are two procedures for importing data, detailed in **Note 7**.

3.5 Calculating Root Length Densities, Root Distances, and Potential Root Extraction Ratio in Soil

After entering the NI per mesh, calculations are made thanks to models embedded inside RACINE2.2 (*see Note 1*). In each base volume corresponding to each counting area (mesh) it is possible to calculate (1) length of the roots per unit volume of soil (RLD cm/cm³), (2) average distance between roots in the soil (ARD, in cm), and (3) PRER (from 0 to 1). The latter is the ratio between the volume of soil potentially available for crop water and nutrient uptake and the corresponding total volume of soil [4, 5]. These three calculations must be done successively (*see Note 8*). More detailed information on the content of these models is available (*see Note 1*).

3.5.1 Calculating Root Length Densities (RLD)

Click on the RLD tab at the bottom of the spreadsheet. A page then appears with, in its lower part, empty RLD tables and, in its upper part, drop-down lists allowing the selection of RLD calculation models from NI. In the three drop-down menus “select a model,” choose the one adapted to the calculation of the root RLD from the available models (depending on type of root and plant, *see Note 1*). The list of available models appears by clicking on the black arrow. The models are chosen for the three types of roots. When the models are selected, click on the “calculate RLD” button. The RLD values are then calculated mesh by mesh and automatically appear in the tables.

3.5.2 Calculating the Average Root Distances (ARD)

Click on the ARD tab to display the ARD page whose configuration is very similar to the RLD page. The procedure is the same as for the RLD calculation, but there is only one model available (*see Note 1*). Click on “calculate ARD” to start the calculation and display the results in the tables (in cm). When there is no root ($RLD = 0$) the table displays “INF” (infinite) which means that the distance between roots is (theoretically) infinite in the volume corresponding to the counting area.

3.5.3 Calculating the Potential Root Extraction Ratio (PRER)

To calculate the PRER, click on the PRER tab at the bottom of the page, which opens the PRER page whose configuration which is very close to the RLD and ARD pages. There is currently only one model available to compute PRER (*see Note 1*). It is necessary to fill in the box: “Max. distance of root uptake (ra)” This corresponds to the estimated maximum distance of migration of water or a nutrient from the soil to a root surface area [2, 4, 5]. After entering this information, click on “Calculate PRER” which leads to the results being displayed in the tables. The values range from 0 (no soil used) to 1 (100% soil used).

3.6 Exporting Results

The data (NI) values entered are saved in RACINE 2.2 as soon as they have been recorded. However, the results of RLD, ARD, and PRER computations are not saved. They are lost as soon as RACINE2.2 is closed (because it is easy and fast to recalculate them). They must therefore be exported before leaving the application. The “Export” tab is used to export the inputs and calculation results to Excel format. By clicking on the “export” tab, a page is displayed that enables to start the export procedure. It is then necessary to specify the place where the file will be stored and its name. The file will be saved in “.xlsx” format. The file is then displayed with the values of NI, RLD, ARD, and PRER in separate tabs as in RACINE2.2. From these tables, now in conventional Excel format, it is possible to arrange them and make statistical calculations and figures. The spreadsheet and tables may appear locked. In such case (*see Note 9*).

3.7 Developing Maps of Root Traits (RLD, ARD, and PRER) Distribution in Soil

It is possible to map the distributions of RLD, ARD, and PRER in a soil profile. In these maps, the previously calculated numerical values are converted into five classes of different intensities of brown color. There are always five classes. This was considered sufficient to show the spatial distribution of root densities. For RLD and PRER, the user can choose the class intervals according to values. For root distances (ARD), it was necessary to set the limits of the five classes, including one reserved for zero RLD values and therefore infinite distances between roots. To create these maps, it is necessary to use two tabs located on the right of the export tab. The first one is “Map preparation” and the second one is “Mapping.” First click on the first one. When clicking on the

“Map preparation” tab, the page that appears is the one shown in Fig. 4, which collects the information required to make a map, it has three main parts:

1. At the top left, there is a table in which the user provides information on: (a) the table size that will collect the data to be mapped: depth, width and size of the square mesh. These characteristics must be the same as those of the table in which the data to be mapped are located, (b) the mini and maxi values of the five classes (*see* **Note 10** to optimize the choices).
2. In the center, there is a table already built, as an example.
3. At the center there are two buttons: “Delete the displayed table” and “Create a new entry table.” A click on the first button deletes the table currently displayed. Clicking on the second button brings up a new table that has the characteristics requested in [1]. It is necessary to fill in table at the top left, then delete the table displayed at the opening of the page, then create a new table. This new table is now identical to the root results table to be mapped. It is thus ready to receive these data. To complete the table in Fig. 4, the simplest way is to copy a table from a RLD or ARD or PRER tab of the application, by a simple copy and paste. It is also possible to enter data directly into the table or to import data from an external folder, provided the table has the right characteristics. The “map preparation” page includes then a completed table. Then, click on the button: “Create a map of RLD” (or ARD or PRER), the “2D maps” tab is then opened. It contains the requested map and class limits. Figure 5 shows an example of an RLD map. The root maps are located in Excel tables. It is therefore possible to modify them in the page itself or in a new page of RACINE2.2 (*see* **Note 11**).

3.8 Models Used for RLD, RD, and PRER Computations

A “models” tab is located at the bottom right of the tab toolbar. In this tab are displayed the models included in RACINE2.2 for calculating RLD, ARD, and PRER. For each of these root traits, there is a two-column table: the name given to the model and the model itself, derived from articles (*see* **Note 1**) and translated into an “Excel” type language (*see* **Note 12**). These models use variables entered or calculated in RACINE2.2 (*see* **Note 12**). For RLD computation from RID, specific models need to be developed for each crop. Published models are available for several crops namely maize, sorghum, sugarcane, rice, eucalyptus, pineapple, pearl millet (*see* **Note 1**). As an example, the simplest model for sorghum [6] is:

$$RLD = Ni/Maille^2 \times 1.97$$

(Maille = Mesh)

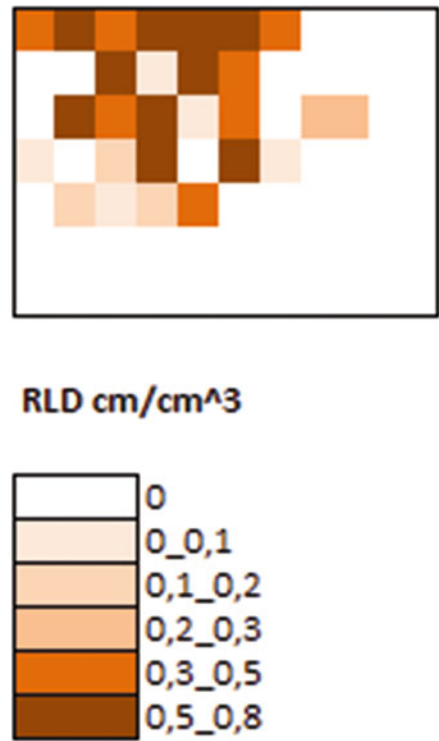


Fig. 5 Example of a root length density distribution map created by RACINE2.2 in the “Mapping” tab and conforming to the class boundaries introduced in the “Map creation” tab

For others crop, as for sugarcane, models may be more complex [7].

There is only one ARD computation model [8]:

$$ARD = (4/\pi)^{0.5}/RLD^{0.5}$$

There is also only one model for calculating the PRER from ARD and *ra* (maximum distance of water or nutrient migration to root), expressed in cm [4, 5]:

$$\text{If } RD/2 \geq ra : PRER = 1.33 \times (ra/ARD)^2, \text{ otherwise : } PRER = 1 - (ARD/(3 \times ra))$$

It is possible to add, modify or delete models (*see* **Notes 12** and **13**).

3.9 Delete Files. Get out RACINE2.2

In the home page (Fig. 1), a button: “open or delete an existing file” is used to delete files that are no longer in use. In this home page, there is also a button “exit” to close the application. It is also possible to exit by closing a page.

4 Notes

1. Root length density (RLD) is not directly related to root counts (NI) on a soil profile. RLD is depending on main roots orientation in soil and their anisotropic degree [9, 10]. So, models calculating RLD from NI are specific for each crop. RACINE2.2 includes models for maize [11], sorghum [6], sugarcane [7], upland rice [12], fine roots of *Eucalyptus grandis* [13], pineapple [14], and pearl millet [15]. Modeling root distances (ARD) is from [8]. PRER approach and modeling are from [2, 4, 5].
2. Under Windows 10 application system, all macros of Excel must be activated before starting RACINE2.2 following the conventional procedure: (a) Open an Excel file, (b) go to "Files Excel Options," "Privacy Management Center," (c) click on "Macro settings," (d) click on "Enable all macros" and on "OK," and (e) close the current Excel file.

When starting for the first time on a workstation, error messages may appear depending on its configuration. In case of an error message: "execution error 339: the tab ctl32 component is not correctly saved", type "cmd" in the "Search" window of the "Start" menu. It then appears at the top of the list "Command prompting," right-click on "Command prompt" and select "Run as administrator," a command window appears. After C:/windows/system32, enter the command "regsvr32 an then c:\windows\system32\tabctl32.ocx", "Enter." If, after this procedure, other error messages appear when the software is restarted, the same operation must be performed by replacing "tabctl32.ocx" with the name of the error entered when RACINE2.2 is opened. For example, if the software indicates: "runtime error 339, the mistifxl.ocx component is not correctly registered," the following command must be entered: "regsvr32 c:\windows\system32\mistifxl.ocx". This operation must be performed for all error messages until the software is working. Then, follow "C:/windows/system32, type regsvr32 comdlg32.ocx" and press "Enter." Click on "start," enter the command, and click on "OK."
3. The application opens by displaying the page displayed at the time of the previous closing. You just have to move to the first page on the left.
4. Existing files do not appear the first starting time. There is a small black arrow. Click in the box, and a second arrow then appears. Click on this second arrow to see the files displayed. Choose one by clicking on it and open it by clicking on "open."
5. Width and depth of the profile must be multiples of the mesh. The planting date must be earlier than the measurement date.

The entries must have dots to separate whole numbers and decimals. All repetitions must have the same characteristics (width, depth, mesh size, planting and measurement dates). If this is not the case, they can no longer be considered as repetitions in the same file and must be included in separate files. The depth is that of the last row of meshes examined. If there are meshes without roots, it is not necessary to enter 0 to indicate this.

6. In tables, each mesh is identified by its center. For example “2.5” of the first column (Fig. 3) corresponds to the first measured mesh with a mesh of 5 cm.
7. The two procedures for importing data are:
 - (a) For files resulting from export by the procedure of RACINE2.2: activate the “Import a RACINE2.2 export” button on the home page (Fig. 1) and follow the procedure. The data import is then automatically done in adapted tables with related information. For the new file to appear, close RACINE2.2 and then restart it.
 - (b) For data entered in other spreadsheet tables, it is necessary to recreate, in RACINE2.2, a file leading to the same table characteristics as those to be imported (same width, depth, mesh). It is then easy to fill in these tables using the copy and paste special: (paste values) process.
8. These three calculations must be done successively because, to calculate ARD, it is necessary to have the RLD corresponding values and to calculate PRER, it is necessary to have the ARD values.
9. The spreadsheet and tables may appear locked, so without the possibility of making calculations or modifications (function bar with greyed-out calculation operations). To unlock it, go to the “File” Table A “Protect files” warning then appears. Click on “remove protection” from each of the tabs and return to the sheet where the results are located.
10. When the table “characteristics and units of the table” and the table containing the data are filled in, the boundaries of the five categories must be chosen to optimize them in relation to the root values. It is possible to choose values for each class limits, ensuring that the maximum value in the table is below the upper limit of the highest value class. For ARD mapping, it is not possible to choose the class limits, as ARD values can theoretically vary from zero to infinity (if the root length is zero). Based on the experience of many studies, five ARD class limits have been selected: “infinite” (no roots in this mesh), “from infinity to 10 cm between the roots,” “from 10 to 5 cm,” “from 5 to 2 cm,” “less than 2 cm.” This last class corresponds to a high root length density.

11. For example, it is possible to separate the map itself and its category scale to change the sizes separately. It is also possible to add depth scales. It is suggested to perform these operations and finalize the formatting in RACINE2.2, then simply copy and save the map as an image in a word processor. It is also possible to copy, without modification, the map into a spreadsheet, but there can then be changes. It is then necessary to edit the map using the functions of this spreadsheet.
12. To write the equation of a model, it is necessary to use allowed parameters, operations and syntax. The allowed parameters, in addition to RLD, ARD, and PRER are: (1) NI: number of intersections measured per mesh, (2) ra: maximum distance (cm) of water or nutrient migration to a root surface, (3) Mesh: length of the sides of the mesh in cm, (4) LAI: Leaf Area Index, (5) Z: depth of measurement in the soil (in cm). The operations and syntax allowed by RACINE2.2 to write model equations are those understood by Microsoft Excel, including conditionality tests. The decimal separator is preferably a dot, but it is possible to choose a coma. It is also possible to contact the authors of the paper to get help with writing a new model.
13. The deletion of a model is done by deleting both the name and the equation, then saving this deletion by the Windows functions: File (top left of the page), then "Save."

References

1. Bohm W (1976) *In situ* estimation of root length at natural soil profiles. J Agric Camb 87:365–368
2. Chopart JL (1999) Relations entre état physique du sol, systèmes racinaires et fonctionnement hydrique du peuplement végétal: outils d'analyse in situ et exemples d'études en milieu tropical à risque climatique élevé. PhD Univ. Grenoble, France 335 p
3. Azevedo M, Chopart JL, Medina de Conti C (2011) Sugarcane root length density and distribution from root intersection counting on a trench-profile. Sci Agric 68:94–101
4. Léfi N, Chopart JL, Roupsard O et al (2011) Genotypic variability of oil palm root system distribution in the field. Consequences for water uptake. Plant Soil 341:505–520
5. Chopart JL, Le Mézo L, Vauclin M (2012) Modelling the potential root water extraction ratio in soil: application to sugar cane on the island of Réunion. In: ISRR (ed) Proceedings of 8th symposium Int. Society root research, Dundee (GB)
6. Chopart JL, Sine B, Dao A, Muller B (2008) Root orientation of four sorghum cultivars: application to estimate root length density from root counts in soil profiles. Plant Root 2:67–75
7. Chopart JL, Rodrigues SR, Azevedo M, Medina C (2008) Estimating sugarcane root length density through root mapping and orientation modelling. Plant Soil 313:101–112
8. Newman J (1966) A method of estimating the total length of root in a sample. J Appl Ecol 3:139–145
9. Lang A, Melhuish F (1970) Lengths and diameters of plant roots in non-random populations by analysis of plane surfaces. Biometrics 26:421–431
10. Van Noordwijk M (1987) Methods for quantification of root distribution pattern and root dynamics in the field. In: Int. Potash Institute Publish (ed) Berne, 20th Cong Int. Potash Institute, Switzerland
11. Chopart JL, Siband P (1999) Development and validation of a model to describe root

- length density of maize from root counts on soil profiles. *Plant Soil* 214:61–74
12. Dusserre J, Audebert A, Radanielson A, Chopart JL (2009) Towards a simple generic model for upland rice root length density estimation from root intersections on soil profile. *Plant Soil* 325:277–288
 13. Maurice J, Laclau JP, Scorzoni D et al (2010) Fine root isotropy in *Eucalyptus grandis* plantations. Towards the prediction of root length densities from root counts on trench walls. *Plant Soil* 334:261–275
 14. Chopart JL, Debaut-Henoque L, Marie-Alphonsine PA et al (2015) Estimating root length density of pineapple (*Ananas comosus* (L.) Merr.) from root counts on soil profiles in Martinique (French West Indies). *Fruits* 70:143–151
 15. Faye A, Sine B, Chopart JL et al (2019) Development of a model estimating root length density from root impacts on a soil profile in pearl millet (*Pennisetum glaucum* (L.) R. Br). Application to measure root system response to water stress in field conditions. *PLoS One* 14 (7):e0214182. <https://doi.org/10.1371/journal.pone.0214182>